

MUHAMMAD HAIDER ZAIDI

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🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, C++, SQL, Bash

Tools/Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, Docker, FastAPI, AWS SageMaker, Kubernetes, Git, Apache Airflow, Hugging Face Transformers, TensorRT, Flask, OpenAI API

Competence: Machine Learning, Deep Learning, MLOps, Computer Vision, Perception, Segmentation, Object Detection, Time Series Prediction, ETL Pipelines, Data Science, Agile Development, Embedded AI, Large Language Models (LLMs), Signal Processing

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated **deep learning models** for **metrological perception tasks**, enhancing **model performance** and **training efficiency** by 15%.
- Explored **novel architectures** and **physics-informed hybrid loss functions** to improve **model accuracy** and **stability** in complex data environments.
- Collaborated with PhD researchers on a **publication** for the FLOMEKO conference, contributing to the scientific community.
- Built **dashboards** using Python for real-time **data visualization** and **model performance monitoring**, ensuring robust system operation.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Optimized vitiligo **segmentation accuracy** to 94% and increased inference speed by 5x using advanced **AI algorithms** and **deep learning models**.
- Developed a **real-time tennis foul detection system** leveraging **TensorRT** for **edge processing**, reducing **compute cost** and **inference time** by 6x.
- Managed end-to-end **MLOps pipelines** on **AWS SageMaker**, proactively addressing **YOLO model data drift** and maintaining **Kubernetes infrastructure**.
- Developed **Proof-of-Concepts (POCs)** using **YOLO**, **few-shot learning**, and **CNNs** for diverse client **perception** and **prediction** challenges.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Developed and integrated **Python-based backend services** for **AI applications**, focusing on robust **data pipelines** and **model interaction**.

- Implemented **RESTful APIs** with Flask to facilitate **AI service deployment** and seamless communication with front-end systems.
- Contributed to an **AI chatbot application** using the **OpenAI API** and relevant **LLM tools**, enhancing natural language understanding.
- Engaged in **agile development practices** and version control with **Git** for collaborative software projects, ensuring timely delivery.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Engineered and deployed **ETL pipelines** for automated **data ingestion** and **cleaning workflows** of structured and unstructured data.
- Developed web scrapers to gather diverse datasets, supporting **client analytics use cases** and **data science initiatives**.
- Optimized data processing routines, improving efficiency and reliability for various client projects.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Architected a weighted-probability **ensemble deep learning model**, integrating DPFunc and DeepGOPlus, leveraging AlphaFold for **feature extraction**.
- Achieved a 4% increase in **model accuracy** through custom **feature engineering** across MF, CC, and BP namespaces, mitigating **taxonomic bias**.

Razanlytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Applied advanced **machine learning models**, including LSTMs and Transformers, for **real-time Bitcoin price forecasting**, achieving competitive **prediction accuracy**.
- Engineered comprehensive features from historical market data and external indicators to significantly enhance **model performance** and robustness for **time series prediction**.

Saaz: AI Self-Playing Chitralli Sitar

2023

Signal Processing, Music Generation

- Collaborated to build the world's first **AI-driven self-playing Chitralli Sitar**, innovatively combining **signal processing** with **music generation LLMs**.
- Showcased advanced **AI application** and creativity, featured in a Coke Studio music video by Hassan Raheem.

Hugging Face

Open-Source

Contributor

- Co-authored the **Zero-shot Image Classification technical documentation** and implementation page on the Hugging Face Hub, facilitating broader adoption.
- Contributed to the **open-source community** by disseminating knowledge on cutting-edge **Transformer-based models** for **perception** tasks.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Pursuing advanced studies in **Artificial Intelligence**, focusing on **deep learning**, **computer vision**, and **reinforcement learning**.
- Engaged in research on **novel ML techniques** for complex problem-solving in real-world applications.

National University of Computer and Emerging Sciences Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Dean's List of Honor
- Completed foundational and advanced coursework in **computer science**, including **data structures**, **algorithms**, and **machine learning fundamentals**.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera**: TensorFlow Developer Certificate
| **Databricks**: Lakehouse Fundamentals | **Datacamp**: Computer Vision in Python